US Patent & Trademark Office
Patent Public Search | Text View

United States Patent 12405164
Kind Code B2
Date of Patent September 02, 2025
Inventor(s) Erickson; Blake et al.

Spatial optical emission spectroscopy for etch uniformity

Abstract

An apparatus includes a base component and a plurality of collimators housed within the base component. Each collimator of the plurality of collimators corresponds to a respective location of a plurality of locations of a wafer in an etch chamber. The plurality of locations includes a center location of the wafer, a plurality of inner ring locations along an inner ring of the wafer associated with a first set of radially symmetric optical emission spectroscopy (OES) signal sampling paths, and a plurality of outer ring locations along an outer ring of the wafer associated with a second set of radially symmetric OES signal sampling paths.

Inventors: **Erickson; Blake (Gilroy, CA), Berding; Keith (Truckee, CA), Kutney; Michael (Santa Clara, CA), Zhu; Zhaozhao (Milpitas, CA), Wu; Tsung Feng (San Jose, CA), Willwerth; Michael D. (Campbell, CA), Ludwig; Jeffrey (San Jose, CA)**

Applicant: **Applied Materials, Inc.** (Santa Clara, CA)

Family ID: **83602247**

Assignee: **Applied Materials, Inc. (Santa Clara, CA)**

Appl. No.: **18/138914**

Filed: **April 25, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20230258500 A1	Aug. 17, 2023

Related U.S. Application Data

continuation parent-doc US 17234940 20210420 US 11668602 child-doc US 18138914

Publication Classification

Int. Cl.: **G01J3/32** (20060101); **G01J3/02** (20060101); **G01J3/443** (20060101); **G01N21/21** (20060101); **G01N21/25** (20060101); **G01N21/68** (20060101); **G01N21/84** (20060101); **G01N21/95** (20060101)

U.S. Cl.:

CPC **G01J3/32** (20130101); **G01J3/0291** (20130101); **G01N21/255** (20130101); **G01N21/68** (20130101); **G01N21/9501** (20130101); G01J3/443 (20130101); G01N21/211 (20130101); G01N2021/8416 (20130101)

Field of Classification Search

CPC: G01J (3/32); G01J (3/0291); G01J (3/443); G01N (21/255); G01N (2021/8416); G01N (21/9501); G01N (21/68); G01N (21/211)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11668602	12/2022	Erickson	356/326	G01J 3/32
2003/0236642	12/2002	Timans	702/99	G01J 5/08
2008/0099435	12/2007	Grimbergen	216/60	H01J 37/32963
2008/0099436	12/2007	Grimbergen	216/60	H01J 37/32963
2008/0176149	12/2007	Grimbergen	216/60	C23F 4/00
2013/0157387	12/2012	Chen	156/345.25	H01L 21/31116
2015/0168130	12/2014	Matsudo	374/161	G01B 9/02021
2017/0314991	12/2016	Meng	N/A	G01N 21/68
2021/0125814	12/2020	Sato	N/A	H01J 37/3222

Primary Examiner: Bologna; Dominic J

Attorney, Agent or Firm: Lowenstein Sandler LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) The present application is a continuation of U.S. patent application Ser. No. 17/234,940, filed on Apr. 20, 2021, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD
(1) The present disclosure relates generally to electronic device fabrication, and, more particularly, relate to spatial optical emission spectroscopy (OES) for etch uniformity.
BACKGROUND
(2) Manufacturing systems produce products based on manufacturing parameters. For example, substrate processing systems produce substrates based on the many parameters of process recipes. Products have performance data based on what parameters were used during production. Etch

process equipment can be used to remove material from areas of a substrate through, e.g., chemical reaction and/or physical bombardment. For example, vacuum etch processes can use plasma to generate gas-phase reactants. During etch processing, an etch rate refers to a rate of material removal, and etch selectivity refers to a ratio of etch rates observed in two materials. There can be a plurality of materials of interest during etch. Such materials include: (1) a target material to be etched; (2) a material underneath the target material; (3) a mask material; and (4) adjacent material to the target material that may be exposed to the etch processing (e.g., process gas).

SUMMARY

(3) The following is a simplified summary of the disclosure in order to provide a basic understanding of some aspects of the disclosure. This summary is not an extensive overview of the disclosure. It is intended to neither identify key or critical elements of the disclosure, nor delineate any scope of the particular implementations of the disclosure or any scope of the claims. Its sole purpose is to present some concepts of the disclosure in a simplified form as a prelude to the more detailed description that is presented later.

(4) In an aspect of the disclosure, an apparatus includes a base component and a plurality of collimators housed within the base component. The plurality of collimators corresponds to a plurality of collection cylinders for sampling optical emission spectroscopy (OES) signals with respect to a plurality of locations of a wafer in an etch chamber. The apparatus further includes a guide, operatively coupled to the plurality of collimators, to guide the sampling of the plurality of OES signals along a plurality of paths for sampling the plurality of OES signals.

(5) In another aspect of the disclosure, a system includes a memory, and a processing device operatively coupled to the memory. The processing device is to facilitate an etch rate uniformity monitoring processing by performing a plurality of operations including initiating an iteration of an etch process to etch a wafer using an etch recipe, receiving an analysis of a plurality of optical emission spectroscopy (OES) signals sampled during the iteration of the etch process with respect to a plurality of locations of the wafer, and performing one or more actions related to etch uniformity based on the analysis of the plurality of OES signals.

(6) In yet another aspect of the disclosure, a method includes initiating, by a processing device, an iteration of an etch process to etch a wafer using an etch recipe, receiving, by the processing device, an analysis of a plurality of optical emission spectroscopy (OES) signals sampled during the iteration of the etch process with respect to a plurality of locations of the wafer, and performing, by the processing device, one or more actions related to etch uniformity based on the analysis of the plurality of OES signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings in which like references indicate similar elements. It should be noted that different references to “an” or “one” embodiment in this disclosure are not necessarily to the same embodiment, and such references mean at least one.

(2) FIG. **1** is a block diagram illustrating a high-level overview of a spatial optical emission spectroscopy (OES) system for etch uniformity monitoring, according to some embodiments.

(3) FIG. **2** is a diagram of a portion of a spatial optical emission spectroscopy (OES) sampling system, according to some embodiments.

(4) FIG. **3** is a diagram illustrating a spatial optical emission spectroscopy (OES) sampling apparatus of a spatial OES sampling system, according to some embodiments.

(5) FIGS. **4**A-**4**B are diagrams of a guide used within a spatial sampling apparatus, according to some embodiments.

(6) FIG. **5** is a diagram of a base component of a spatial optical emission spectroscopy (OES) sampling apparatus, according to some embodiments.
(7) FIG. **6** is a diagram of an etch uniformity monitoring sub-system, according to some embodiments.
(8) FIG. **7** is a flow diagram of a method to implement spatial optical emission spectroscopy (OES) for etch uniformity, according to some embodiments.
(9) FIG. **8** is a block diagram illustrating a computer system, according to some embodiments.

DETAILED DESCRIPTION

(10) Embodiments described herein relate to optical emission spectroscopy (OES) for etch uniformity. As etch process recipes are being developed to achieve a desired etch pattern, methods of obtaining feedback are limited. For example, methods of obtaining feedback include: (1) external metrology with wafers; (2) external metrology with coupons; (3) single point in-situ reflectometry; and (4) ellipsometry.
(11) To perform external metrology with wafers, wafers with films of materials of interest can be premeasured with an ellipsometer or other external metrology. Wafers can be etched with a proposed etch process and then re-measured with external metrology. Etch selectivity can then be determined by comparing wafers with films of different materials.
(12) To perform external metrology with coupons, similar to external metrology with wafers, coupons with films of materials of interest are placed on a wafer. Each coupon can be baselined for film thickness with an ellipsometer or other external metrology. Selectivity can be determined by comparing etch rates of each material.
(13) To perform single point in-situ reflectometry, a wafer or coupon on a carrier is measured in-situ with single point reflectometry, which is used to monitor etch rate during the etch process. Etch selectivity can then be determined by measuring each coupon/wafer separately. This method can require one wafer or one coupon on a wafer, and one process per material.
(14) With respect to ellipsometry, an in-situ ellipsometer can be added onto the etch chamber to measure a spot adjacent to the spot measured by reflectometry. The ellipsometer can be bulky and difficult to align, can require chamber modification, and the measurement method is different and must be calibrated. It is practically limited to allow only one additional measurement point.
(15) OES can provide valuable information about the gases present and chemical byproducts generated during a plasma etch process. Etch chambers can be outfitted with an optical signal collection device coupled to a spectrometer via optical channels (e.g., fiber optic cables). The spectrometer can identify specific wavelength photons that are uniquely emitted from specific materials when excited in a radio frequency (RF) generated plasma. A port or window may be located on the side of the etch chamber to sample the global optical emission from the chamber. However, this provides little to no information regarding the spatial intensity of the emission.
(16) To address these and other limitations, the embodiments described herein provide for a system that permits OES spatial sampling within an etch chamber. For example, some embodiments described herein can use multiple collimators to strategically sample photons from various paths in an etch chamber to identify spatial variations during an etch process using an etch recipe that can impact etch uniformity on a wafer. In addition, some embodiments described herein can develop and/or optimize the etch recipe for etch uniformity and/or for monitoring etch uniformity drifts as the etch chamber changes over RF hours.
(17) The spatial OES system can include a spatial OES sampling apparatus. In some embodiments, the spatial OES sampling apparatus can include a number of collimators each corresponding to a collection cylinder for sampling OES signals along different paths above a wafer in the etch chamber. In some embodiments, the collimators can be positioned above the top/center window and point to particular locations above the wafer. For example, the locations above the wafer can include a center of the wafer, locations along an inner ring of the wafer (referred to herein as the "wafer middle"), and locations along an outer ring of the wafer (referred to herein as the "wafer

edge”). Any suitable number of sampling paths can be obtained in accordance with the embodiments described herein. For example, in some embodiments, 7 collimators corresponding to 7 collection cylinders can be provided, with one collection cylinder directed to the wafer center, three collection cylinders directed to the wafer middle, and three collection cylinders directed to the wafer edge. The collimators may not need to be positioned above the top/center window, and can be positioned in any suitable configuration in accordance with the embodiments described herein.
(18) The spatial OES system can further include an etch uniformity monitoring sub-system. Each of the OES signals collected by the spatial OES sampling apparatus can be transmitted by a corresponding optical channel (e.g., fiber optic cable) from each of the collimators to the etch uniformity monitoring sub-system for monitoring etch uniformity. More specifically, the etch uniformity monitoring sub-system can include at least one optical detector (e.g., spectrometer) for performing an optical analysis (e.g., spectroscopic analysis) on the OES signals to analyze, e.g., intensity and/or wavelength. For example, each of the OES signals can be sent to a switch device (e.g., a multiplexer) to route each of the OES signals to a single optical detector. As another example, each of the OES signals can be sent to its own optical detector, without requiring a switch device. The results of the optical analysis can be used by a processing device of the etch uniformity monitoring sub-system to further develop and/or optimize an etch recipe for etch uniformity.
(19) Embodiments described herein advantageously overcome the limitations of other measurement techniques by providing “in-situ” feedback on etch uniformity with correlation to external metrology. Additionally, time to develop etch recipes in the lab can be improved.
(20) FIG. **1** is a diagram of a high-level overview of a spatial optical emission spectroscopy (OES) system **100** for etch uniformity. More specifically, as will be described in further detail below, the system **100** can implement spatial OES with multiple OES signals to analyze etch uniformity with respect to a plasma etch process performed in an etch chamber, and can be used to develop and/or optimize an etch recipe based on results of the analysis. As shown, the system **100** includes a wafer **110**, a spatial OES sampling apparatus (“apparatus”) **120** and an etch uniformity monitoring sub-system (“sub-system”) **130**. The wafer **110** can be placed on an electrostatic chuck or other suitable apparatus (not shown) to secure the wafer **110** within the etch chamber.
(21) As will be described in further detail below with reference to FIG. **2**, the apparatus **120** supports a number of collimators (not shown) arranged in a variety of directions and/or angles to perform spatial OES sampling at respective locations on the wafer **110** during the iteration of the etch process. More specifically, each of the collimators corresponds to a collection cylinder, such as the collection cylinder **122**, for sampling OES signals at different locations above the wafer **110** each corresponding to a sampling point. As will be described in further detail below with reference to FIGS. **2-3**, the collimators can be arranged with a base component (not shown) of the apparatus **120** with respect to different radii of the wafer **110**. In some embodiments, the apparatus **120** supports 7 collimators corresponding to 7 collection cylinders. As will be further described below with reference to FIGS. **2-4**, the apparatus **120** can further include a guide (not shown) that is designed to guide the sampling of the OES signals along the respective paths for collecting a number of different OES zones (e.g., **7** OES zones).
(22) The sub-system **130** includes at least one optical detector **132** operatively coupled to a switch device **134**, a processing device **136** operatively coupled to the components **132** and **134**, and a memory device **138** operatively coupled to the processing device **136**. In some embodiments, the optical detector **132** is a spectrometer.
(23) The processing device **136** can be used to initiate an iteration of an etch process using an etch recipe **139** stored in the memory **138**. More specifically, the etch recipe **139** can be a plasma etch recipe. The etch recipe **139** has etch recipe conditions (e.g., gas type, gas concentration, pressure, power, pulsing, etc.) that control the etch effect on the materials during the iteration of the etch process initiated by the processing device **136**.
(24) The OES signals sampled during the iteration of the etch process can be routed via optical

channels (e.g., a bundle of fiber optic cables) to the switch device **134** to distribute and route the OES signals to the optical detector **132**. In some embodiments, the switch device is a multiplexer. More specifically, the switch device **134** can switch individual optical channels (e.g., fibers) to be sampled by the optical detector **132** in series for performing spectroscopy on the OES signals. However, in alternative embodiments, a number of optical detectors can be used in place of the switch device **134**, such that each optical channel is coupled to a corresponding optical detector.
(25) The processing device **136** can further analyze the OES signals sampled by the collection cylinders (e.g., collection cylinder **122**) to monitor and/or control etch uniformity. More specifically, the etch rate can be measured by the processing device **136** using a model based on the OES signals. The processing device **136** can then optimize and/or maintain etch uniformity based on the measurements (e.g., using suitable machine learning techniques), which can be used to further develop and/or optimize the etch recipe **139** for etch uniformity.
(26) FIG. **2** illustrates a portion of a spatial optical emission spectroscopy (OES) sampling system ("system") **200**, according to some embodiments. More specifically, the system **200** can be implemented within an etch chamber to monitor etch uniformity. As shown, the system includes a wafer **210** being etched and a spatial OES sampling apparatus ("apparatus") **220**.
(27) The wafer **210** can include any suitable material that can be etched by a plasma etch process. Examples of materials include, but are not limited to, oxides, nitrides, polysilicon, and photoresists. As show, the wafer **210** includes an inner ring **212** corresponding to the wafer middle and an outer ring **214** corresponding to the wafer edge. More specifically, the outer ring **214** can be as far from the wafer center as possible to perform spatial OES sampling.
(28) The apparatus **220** includes a base component **222** housing a number collimators including collimators **224-1** through **224-3**, and a guide **226**. Each of the collimators corresponds to a collection cylinder. For example, the collimator **224-1** corresponds to a collection cylinder **228-1**, the collimator **224-2** corresponds to a collection cylinder **228-2**, and the collimator **224-3** corresponds to a collection cylinder **228-3**. Another collimator (not shown) corresponds to a collection cylinder **228-4**.
(29) The guide **226** is designed to point the collection cylinders to respective locations on the wafer **210** for spatial OES sampling. For example, if there are 7 collimators configured to generate 7 collection cylinders, one of the collection cylinders can be pointed to the center of the wafer **210**, three of the collection cylinders can be pointed to respective locations along the outer ring **212** and three of the collection cylinders can be pointed to respective locations along the inner ring **214**. For example, in this illustrative example, the collection cylinder **228-1** is pointed to the center of the wafer **210**, the collection cylinders **228-2** and **228-4** are pointed to respective locations along the inner ring **214**, and collection cylinders **228-3** and **228-5** are pointed to respective locations along the outer ring **212**. It may be desirable for this group of 3 collection cylinders to provide a similar result, as this means that the impact of the process on the wafer **210** is symmetric around the center of the wafer **210**. If the behavior is not similar, it indicates skew. Thus, by having 3 collection cylinders in the wafer middle and the wafer edge, skew can be assessed.
(30) In some embodiments, the guide **226** has a length of between about 20 mm and about 30 mm. More specifically, the guide **226** can have a length of, e.g., about 25.7 mm.
(31) Near the guide **226**, each of the collection cylinders pass through a similar zone, but diverge as they go toward their respective locations above the wafer **210**. The distance between the guide **226** and the wafer **210**, "L1," can be determined based on the desired radii/diameters of the outer ring **212** and the inner ring **214**, as well as the respective locations that the collimators are pointed to along the outer ring **212** and the inner ring **214**. In some embodiments, L1 is between about 147 mm and about 157 mm. More specifically, L1 can be, e.g., about 152.4 mm (6.00 inches). In some embodiments, a distance between the collection cylinder **228-1** and the collection cylinder **228-3**, "L2," is between about 120 mm and about 130 mm, and a distance between the collection cylinder **228-1** and the collection cylinder **228-2**, "L3," is between about 63 mm and about 73 mm. More

specifically, L2 can be, e.g., about 125 mm and L3 can be, e.g., about 68 mm. In some embodiments, an angle between the collection cylinder **228-1** and the collection cylinder **228-3**, “A1,” is between about 30° and about 40°, and an angle between the collection cylinder **228-1** and the collection cylinder **228-2**, “A2,” is between about 16° and about 26°. More specifically A1 can be, e.g., about 35° and A2 can be, e.g., about 21°.

(32) If the etch process is more aggressive in one radial zone on the wafer than another, then a higher concentration of byproducts may be in the plasma and more photons would be collected by the collimator pointed in that direction. By normalizing the data relative to a starting condition, non-uniformity due to other factors such as channel (e.g., fiber) transmission or alignment can be subtracted out.

(33) The system **200** can detect OES spatial differences due to etch recipe inputs. More specifically, as will be described in further detail below with reference to FIGS. **3-4**, a number of scallops can be built into the guide **226** for guiding the sampling of the OES signals along the respective paths with respect to the outer ring **212** and the inner ring **214**. The collection cylinder **228-1** can be directed above the center of the wafer **210** without a scallop through the opening of the guide **226**. Moreover, the apparatus **200** can include an optical port or window (not shown) to allow the multiple collimators (e.g., 7 collimators) to collect the optical signals from the various locations above the wafer **210**.

(34) Due to the geometric limitations in the collection cylinders, it may be observed that the “edge” pointed cylinders are longer than the “center” cylinder, which may have an impact on the relationship between etch rate and sensor signals. This difference can be accounted for using machine learning through system training.

(35) FIG. **3** illustrates a spatial OES sampling apparatus (“apparatus”) **300** of a spatial OES sampling system, according to some embodiments. The apparatus **300** is similar to the apparatus **200** described above with reference to FIG. **2**.

(36) As shown, the apparatus **300** includes an optical port **310**, a base component **320**, a number of collimators including a collimator **330**, a guide **340**, and multiple collection cylinders corresponding to respective ones of the multiple collimators, including a collection cylinder **350** corresponding to the collimator **330**. As described above with reference to FIG. **2**, the optical port **310** can be a window to allow the multiple collimators (e.g., 7 collimators) to collect optical signals along various paths with respect to the wafer (e.g., the wafer center, an inner ring corresponding to the wafer middle, and an outer ring corresponding to the wafer edge). That is, the multiple collection cylinders pass through the optical port **310** at different angles to capture photons for OES analysis. As mentioned above with reference to FIG. **2** and as will be described in further detail below with reference to FIGS. **4**A and **4**B, respective scallops built into the guide **340** can be used to guide the sampling of the OES signals along the respective paths with respect to the wafer.

(37) FIGS. **4**A and **4**B illustrate respective perspective views of a guide **400**, according to some embodiments. The guide **400** can be used within a spatial OES sampling apparatus, as described above with reference to FIGS. **1-3**. As shown, the guide **400** has a top surface **410**, a bottom surface **420**, a number of scallops **430**, and an opening **440**. The scallops **430** guide the sampling of the OES signals along respective paths with respect to a wafer. In this illustrative example, the guide **400** has 6 scallops **430**, such that the guide **400** can support 7 collection cylinders for enabling collection of optical signals from **7** different OES zones.

(38) FIG. **5** illustrates an example base component **500** of a spatial OES sampling apparatus (“apparatus”), such as the apparatus **200** of FIG. **2** and the apparatus **300** of FIG. **3**. As shown, the base component **500** includes an upper clamp **510**, a base **520**, and multiple ports including port **530**. As further shown in this example, a set of optical channels **540** including an optical channel **542** is secured to the upper clamp **510**. Collimators are attached to respective ends of the optical channels of the set **540**, such as a collimator **544** attached to an end of the optical channel **542**. Each collimator is inserted into a respective port (e.g., the collimator **544** is inserted into the port

530). As described herein above, each collimator corresponds to a collection cylinder for sampling OES signals from a corresponding location above a wafer. In this illustrative example, there are 7 ports and 7 collimators, with one collimator corresponding to a collection cylinder located above the center of the wafer, three collimators corresponding to three collection cylinders located at respectively positions above an inner ring of the wafer (the wafer middle), and three collimators corresponding to three collection cylinders located at respectively positions above an outer ring of the wafer (the wafer edge).

(39) FIG. **6** is a diagram of a system **600** including an etch uniformity monitoring sub-system ("sub-system"), according to some embodiments. As shown, the system **600** includes an etch chamber **610**, a switch device **620** (e.g., a multiplexer), and an optical detector **630** (e.g., spectrometer).

(40) A number of collimators **612** are operatively coupled to the etch chamber **610**. Each collimator corresponds to a collection cylinder for collecting optical emission signals ("signals") **614**, and the number of collimators **612** can be equal to the number of signal measurement locations with respect to a wafer in the etch chamber **610**. For example, the collimators **612** can be located above the wafer to collect signals **614** from respective locations above the wafer. In this illustrative embodiment, there are seven collimators corresponding to seven locations with respect to the wafer. The signals **614** can be sent to a switch device **620** via respective optical channels **615** (e.g., fiber optic cables). The switch device **620** can select one of the signals **614** at a time as a selected signal **622** to be sent to the optical detector **630** via an optical channel **625** for analysis. In alternative embodiment, instead of having the switch device **620**, each of the signals **614** can be sent to a respective optical detector.

(41) FIG. **7** is a flow diagram of a method **700** to implement spatial optical emission spectroscopy (OES) for etch uniformity, according to some embodiments. The method **700** can be performed by processing logic that can include hardware (e.g., processing device, circuitry, dedicated logic, programmable logic, microcode, hardware of a device, integrated circuit, etc.), software (e.g., instructions run or executed on a processing device), or a combination thereof. In some embodiments, the method **700** is performed by the processing device **136** of FIG. **1**. Although shown in a particular sequence or order, unless otherwise specified, the order of the processes can be modified. Thus, the illustrated embodiments should be understood only as examples, and the illustrated processes can be performed in a different order, and some processes can be performed in parallel. Additionally, one or more processes can be omitted in various embodiments. Thus, not all processes are required in every embodiment. Other process flows are possible.

(42) At block **710**, the processing logic initiates an iteration of an etch process to etch a wafer using an etch recipe. In some embodiments, the processing logic can initiate the iteration of the etch process in response to receiving a request to initiate the iteration of the etch process. For example, the processing logic, after receiving the request, can cause the exposure of the multiple materials to etchants as determined by the etch recipe. The thickness of one or more of the materials can decrease over time as a result of the etch recipe being used.

(43) The etch process can be a plasma etch process. The etch recipe has etch recipe conditions (e.g., gas type, gas concentration, pressure, power, pulsing, etc.) that control the etch effect on the materials during the iteration of the etch process.

(44) At block **720**, processing logic receives an analysis of multiple optical signals corresponding to respective locations of the wafer during the iteration of the etch process. More specifically, the multiple optical signals can be optical emission spectroscopy (OES) signals. For example, the analysis can determine etch rates with respect to each location. In some embodiments, multiple collection cylinders can be pointed at the respective locations of the wafer for sampling respective ones of the optical signals, with each of the collection cylinders corresponding to a collimator attached to an optical channel (e.g., a fiber optic cable). An optical channel (e.g., fiber optic cable) connected to each collimator can be used to transmit an optical signal to an optical detector (e.g.,

spectrometer) to perform the analysis (e.g., spectroscopy). In some embodiments, the optical signals are routed to a single optical detector using a switch device (e.g., multiplexer). In other embodiments, each optical signal can be directly routed to a corresponding optical detector.

(45) At block **730**, the processing logic performs one or more actions related to etch uniformity based on the analysis of the optical signals. For example, etch rates determined by the analysis can be used to measure etch uniformity with respect to the various locations of the wafer. For example, radial and/or skew etch uniformity can be measured.

(46) The one or more actions can include optimizing the etch recipe for etch uniformity. For example, the processing logic can implement any suitable machine learning technique to determine which output results in a uniform etch process, and then etch recipe options can be developed to achieve etch uniformity. Accordingly, etch recipe quality can be improved.

(47) The one or more actions can include monitoring etch uniformity. For example, etch uniformity can be monitored by the processing logic as a function of RF hours. In some embodiments, a baseline of what is uniform in production can be established, and the etch uniformity can be monitored to identify the existence of etch uniformity drift.

(48) The one or more actions can include controlling etch uniformity. For example, the processing logic can control etch uniformity by changing etch recipe inputs to achieve desired OES outputs (e.g., OES intensity). This can be implemented by an automatic control system for maintaining etch rates and etch uniformity by changing equipment inputs.

(49) The method **700** can be used to analyze the effect that different environmental conditions have on OES parameters with respect to the center, middle, and edge locations of a wafer being etched. For example, the method **700** can be used to analyze the effect of RF ratio (Ra) on OES intensity with respect to the center, middle and edge locations of a wafer being etched. In some implementations, the method **700** can be used to analyze the effect of Ra between inner and outer coils on OES intensity using 7 collection cylinders. In this case, the **3** middle ring points, and 3 edge ring points can be averaged to provide one value for center, middle, edge. It can be shown that higher Ra can result in lower OES signal for all locations, higher Ra can cause higher change in the center than the edge, and as Ra increases, edge intensity can increase. As another example, the method **700** can be used to analyze the effect of pressure on OES intensity with respect to the center, middle and edge locations of a wafer being etched. It can be shown that pressure decreases OES intensity but impacts the center more than the edge.

(50) The method **700** can be further used to analyze etch rate as a function of OES parameters with respect to the center, middle and edge locations of a wafer being etched. For example, the spatial OES method can be used to analyze normalized etch rate as a function of normalized OES intensity with respect to the center, middle and edge locations of a wafer being etched. It can be shown that there is a direct relationship between OES intensity and etch rate, and that etch rate can be highest at the edge locations and lowest at the center location. Further details regarding the method **700** are described above with reference to FIGS. **1-6**.

(51) FIG. **8** is a block diagram illustrating a computer system **800**, according to certain embodiments. In some embodiments, the computer system **800** is one or more of client device or server.

(52) In some embodiments, computer system **800** is connected (e.g., via a network, such as a Local Area Network (LAN), an intranet, an extranet, or the Internet) to other computer systems. In some embodiments, computer system **800** operates in the capacity of a server or a client computer in a client-server environment, or as a peer computer in a peer-to-peer or distributed network environment. In some embodiments, computer system **600** is provided by a personal computer (PC), a tablet PC, a Set-Top Box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, switch or bridge, or any device capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that device. Further, the term “computer” shall include any collection of computers that individually or jointly execute a set

(or multiple sets) of instructions to perform any one or more of the methods described herein.
(53) In a further aspect, the computer system **800** includes a processing device **802**, a volatile memory **804** (e.g., Random Access Memory (RAM)), a non-volatile memory **806** (e.g., Read-Only Memory (ROM) or Electrically-Erasable Programmable ROM (EEPROM)), and a data storage device **816**, which communicate with each other via a bus **808**.
(54) In some embodiments, processing device **802** is provided by one or more processors such as a general purpose processor (such as, for example, a Complex Instruction Set Computing (CISC) microprocessor, a Reduced Instruction Set Computing (RISC) microprocessor, a Very Long Instruction Word (VLIW) microprocessor, a microprocessor implementing other types of instruction sets, or a microprocessor implementing a combination of types of instruction sets) or a specialized processor (such as, for example, an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Digital Signal Processor (DSP), or a network processor).
(55) In some embodiments, computer system **800** further includes a network interface device **822** (e.g., coupled to network **874**). In some embodiments, computer system **800** also includes a video display unit **810** (e.g., an LCD), an alphanumeric input device **812** (e.g., a keyboard), a cursor control device **814** (e.g., a mouse), and a signal generation device **820**.
(56) In some implementations, data storage device **816** includes a non-transitory computer-readable storage medium **824** on which store instructions **826** encoding any one or more of the methods or functions described herein. For example, the instructions **826** can include the etch recipe component **139** of FIG. **1**, which, when executed, can implement a method for etch uniformity monitoring, such as the method **700** of FIG. **7**.
(57) In some embodiments, instructions **826** also reside, completely or partially, within volatile memory **804** and/or within processing device **802** during execution thereof by computer system **800**, hence, in some embodiments, volatile memory **804** and processing device **802** also constitute machine-readable storage media.
(58) While computer-readable storage medium **824** is shown in the illustrative examples as a single medium, the term “computer-readable storage medium” shall include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of executable instructions. The term “computer-readable storage medium” shall also include any tangible medium that is capable of storing or encoding a set of instructions for execution by a computer that cause the computer to perform any one or more of the methods described herein. The term “computer-readable storage medium” shall include, but not be limited to, solid-state memories, optical media, and magnetic media.
(59) In some embodiments, the methods, components, and features described herein are implemented by discrete hardware components or are integrated in the functionality of other hardware components such as ASICS, FPGAs, DSPs or similar devices. In some embodiments, the methods, components, and features are implemented by firmware modules or functional circuitry within hardware devices. In some embodiments, the methods, components, and features are implemented in any combination of hardware devices and computer program components, or in computer programs.
(60) Unless specifically stated otherwise, terms such as “training,” “identifying,” “further training,” “re-training,” “causing,” “receiving,” “providing,” “obtaining,” “optimizing,” “determining,” “updating,” “initializing,” “generating,” “adding,” or the like, refer to actions and processes performed or implemented by computer systems that manipulates and transforms data represented as physical (electronic) quantities within the computer system registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices. In some embodiments, the terms “first,” “second,” “third,” “fourth,” etc. as used herein are meant as labels to distinguish among different elements and do not have an ordinal meaning according to their numerical designation.

(61) Examples described herein also relate to an apparatus for performing the methods described herein. In some embodiments, this apparatus is specially constructed for performing the methods described herein, or includes a general purpose computer system selectively programmed by a computer program stored in the computer system. Such a computer program is stored in a computer-readable tangible storage medium.
(62) The methods and illustrative examples described herein are not inherently related to any particular computer or other apparatus. In some embodiments, various general purpose systems are used in accordance with the teachings described herein. In some embodiments, a more specialized apparatus is constructed to perform methods described herein and/or each of their individual functions, routines, subroutines, or operations. Examples of the structure for a variety of these systems are set forth in the description above.
(63) The above description is intended to be illustrative, and not restrictive. Although the present disclosure has been described with references to specific illustrative examples and implementations, it will be recognized that the present disclosure is not limited to the examples and implementations described. The scope of the disclosure should be determined with reference to the following claims, along with the full scope of equivalents to which the claims are entitled.

Claims

1. An apparatus comprising: a base component; and a plurality of collimators housed within the base component, each collimator of the plurality of collimators corresponding to a respective location of a plurality of locations of a wafer in an etch chamber, wherein the plurality of locations comprises a center location of the wafer, a plurality of inner ring locations along an inner ring of the wafer associated with a first set of radially symmetric optical emission spectroscopy (OES) signal sampling paths, and a plurality of outer ring locations along an outer ring of the wafer associated with a second set of radially symmetric OES signal sampling paths.
2. The apparatus of claim 1, further comprising a guide, operatively coupled to the plurality of collimators, to guide each OES signal of the plurality of OES signals along a respective spatial OES signal sampling path.
3. The apparatus of claim 1, wherein each collimator of the plurality of collimators corresponds to a respective collection cylinder of a plurality of collection cylinders, and wherein each collection cylinder of the plurality of collection cylinders is to collect a respective OES signal of a plurality of OES signals.
4. The apparatus of claim 1, wherein the inner ring comprises 3 inner ring locations and the outer ring comprises 3 outer ring locations, and the plurality of collimators comprises 7 collimators.
5. The apparatus of claim 1, wherein the base component further comprises: an upper clamp for securing a set of optical channels, wherein each collimator of the plurality of collimators is attached to a respective end of an optical channel of the set; a base; and a plurality of ports formed within the base, wherein each collimator of the plurality of collimators is inserted into a respective port of the plurality of ports.
6. A system comprising: a memory; and a processing device, operatively coupled to the memory, to perform operations comprising: initiating an iteration of an etch process to etch a wafer using an etch recipe; receiving, from an optical detector, a spatial optical emission spectroscopy (OES) analysis of a plurality of OES signals, wherein each OES signal of the plurality of OES signals corresponds to a respective location of a plurality of locations of the wafer, and wherein the plurality of locations comprises a center location of the wafer, a plurality of inner ring locations along an inner ring of the wafer associated with a first set of radially symmetric OES signal sampling paths, and a plurality of outer ring locations along an outer ring of the wafer associated with a second set of radially symmetric OES signal sampling paths; and performing one or more actions related to etch uniformity based on the spatial OES analysis.

7. The system of claim 6, further comprising a plurality of collimators housed within a base component.
8. The system of claim 7, further comprising a guide, operatively coupled to the plurality of collimators, to guide each OES signal of the plurality of OES signals along a respective spatial OES signal sampling path.
9. The system of claim 7, wherein each collimator of the plurality of collimators corresponds to a respective collection cylinder of a plurality of collection cylinders, and wherein each collection cylinder of the plurality of collection cylinders is to collect a respective OES signal of the plurality of OES signals.
10. The system of claim 7, wherein the base component further comprises: an upper clamp for securing a set of optical channels, wherein each collimator of the plurality of collimators is attached to a respective end of an optical channel of the set; a base; and a plurality of ports formed within the base, wherein each collimator of the plurality of collimators is inserted into a respective port of the plurality of ports.
11. The system of claim 6, wherein the processing device is operatively coupled to a switch device to enable the plurality of OES signals to be selectively analyzed with the optical detector.
12. The system of claim 6, wherein the spatial OES analysis determines etch rates with respect to each of the locations.
13. The system of claim 6, wherein performing the one or more actions comprises optimizing etch uniformity.
14. The system of claim 6, wherein performing the one or more actions comprises monitoring etch uniformity.
15. The system of claim 6, wherein performing the one or more actions comprises controlling etch uniformity.
16. The system of claim 6, wherein the plurality of inner ring locations comprises 3 inner ring locations and the plurality of outer ring locations comprises 3 outer ring locations.
17. A method comprising: initiating, by a processing device, an iteration of an etch process to etch a wafer using an etch recipe; receiving, by the processing device from an optical detector, a spatial optical emission spectroscopy (OES) analysis of a plurality of OES signals, wherein each OES signal of the plurality of OES signals corresponds to a respective location of a plurality of locations of the wafer, and wherein the plurality of locations comprises a center location of the wafer, a plurality of inner ring locations along an inner ring of the wafer associated with a first set of radially symmetric OES signal sampling paths, and a plurality of outer ring locations along an outer ring of the wafer associated with a second set of radially symmetric OES signal sampling paths; and performing, by the processing device, one or more actions related to etch uniformity based on the spatial OES analysis.
18. The method of claim 17, wherein the analysis of the plurality of OES signals determines etch rates with respect to each of the locations.
19. The method of claim 17, wherein performing the one or more actions comprises performing at least one of: optimizing etch uniformity, monitoring the etch uniformity, or controlling the etch uniformity.
20. The method of claim 17, wherein the plurality of inner ring locations comprises 3 inner ring locations and the plurality of outer ring locations comprises 3 outer ring locations.